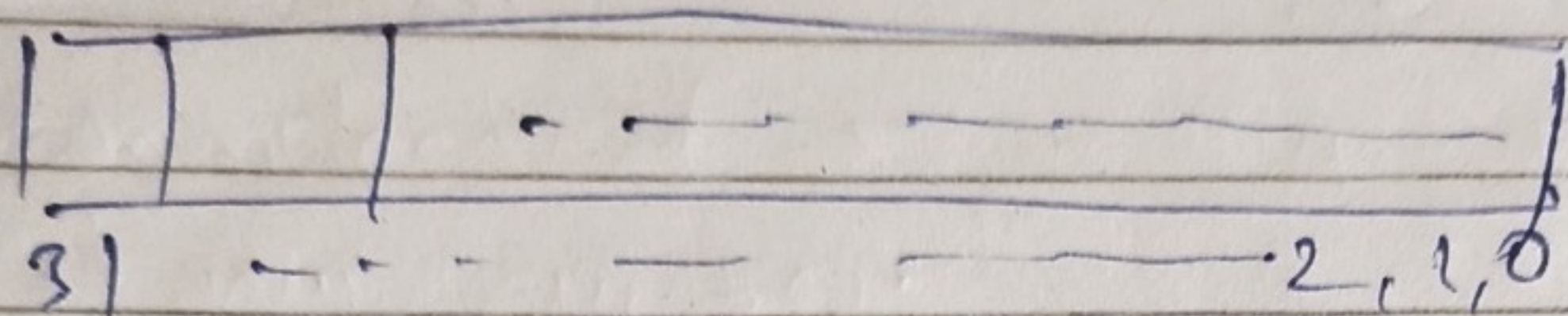


* Bit Manipulation :-

int $\rightarrow$ 4 byte $\times$ 1 byte = 8 bits

then 4 Byte = 32 bits



let 4 bits.

4 $\rightarrow$ $\frac{0/1}{3}$ $\frac{0/1}{2}$ $\frac{0/1}{1}$ $\frac{0/1}{0}$

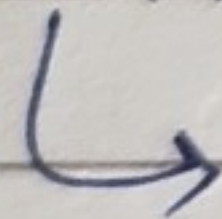
2 2 2 2 $\rightarrow$ bits

$\therefore 2^4 = 16$ bits. (0 - 15)

So, we can store.

$[0, 2^n - 1]$

* MSB & LSB $\rightarrow$ least significant bite.



Most significant bite

For signed,

1/0

MSB

LSB.

For 0 $\rightarrow$ (+)

1 $\rightarrow$ (-)

Incase of signed MSB Pehle se occupied hota hai.

examples:

take 3 bits.

+ve msb	msb	0	0	0	→ +0
		0	0	1	→ +1
		0	1	0	→ +2
		0	1	1	→ +3

-ve msb		1	0	0	→ -0
		1	0	1	→ -1
		1	1	0	→ -2
		1	1	1	→ -3

Note:- Signed number store karne ka drawback
hota hai ki hm usme both (-0) & ~~(+0)~~ (+0)
store kar sakte hai.

Here Range.

$$[-3, 3]$$

$$[-2^{n-1} + 1, (2^{n-1} - 1)] \text{ for signed number.}$$

$$[0, 2^n - 1] \text{ for unsigned number.}$$

Q Add two numbers in binary:-

$$\begin{array}{r}
 \begin{array}{cccccc}
 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\
 1 & 0 & 1 & 1 & 0 & 0 \\
 0 & 1 & 0 & 0 & 1 & 1 \\
 \hline
 1 & 1 & 1 & 1 & 1 & 1
 \end{array}
 \end{array}
 \begin{array}{l}
 \rightarrow 44 \\
 \rightarrow 19 \\
 \hline
 \rightarrow 63
 \end{array}$$

Example:-

$$\begin{array}{r}
 \text{carry} \quad 1 \quad \text{carry} \quad 1 \quad \text{carry} \quad 1 \\
 1 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 1 \\
 + \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\
 \hline
 1 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 1 \\
 \hline
 \end{array}$$

2's complement $\rightarrow$ By this we can represent -ve of any number.

let

$$52 \rightarrow \underline{0} \quad \underline{0} \quad \underline{32} \quad \underline{16} \quad \underline{0} \quad \underline{4} \quad \underline{0} \quad \underline{0}$$

$$\text{or } 52 \rightarrow 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0$$

first we need to create 1's complement $\Rightarrow$ just reverse all numbers.

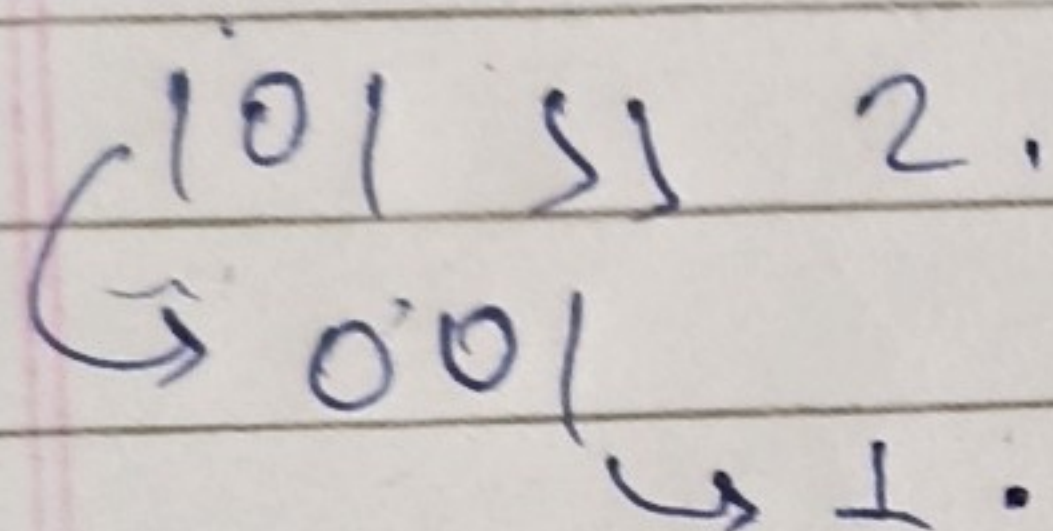
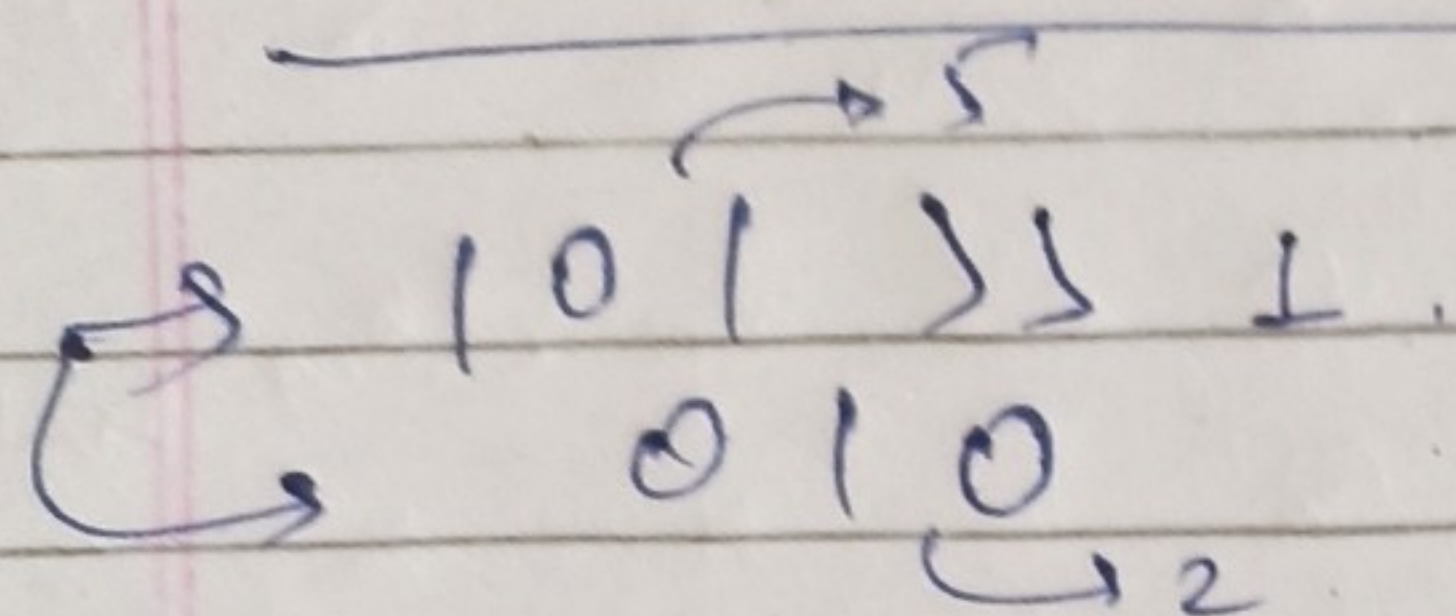
$$1's \text{ C} \rightarrow 52 \rightarrow 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1$$

$$2's \text{ complement} \rightarrow 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 0 = -52.$$

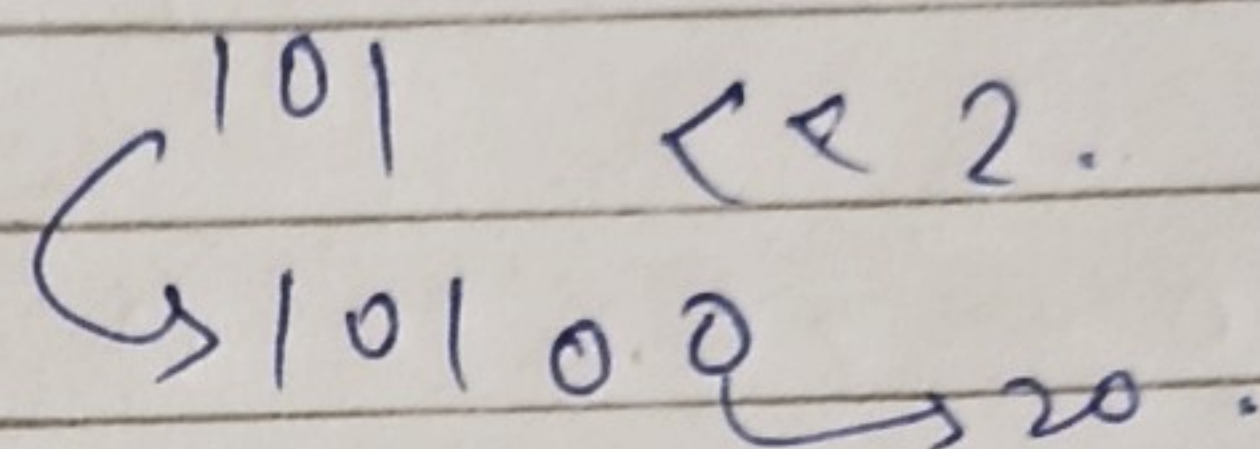
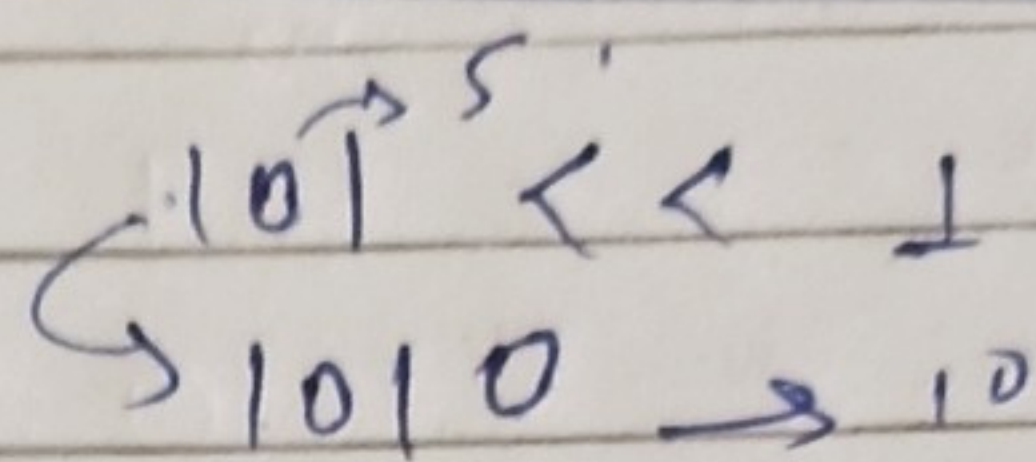
* Bitwise operator:-

a	b	AND	OR	XOR
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

★ Right shift
 $a \gg b$



left shift
 $a \ll b$



Observation.

$$5 \gg 1 \rightarrow 2$$

$$5 \gg 2 \rightarrow 1$$

$$5 \ll 1 \rightarrow 10$$

$$5 \ll 2 \rightarrow 20$$